AquaSentry Home: Hardware Assembly Manual

Complete Step-by-Step Guide to Building the In-Tank Water Quality & Arsenic Surveillance Sentry

Executive Overview

This manual provides complete, beginner-friendly, step-by-step instructions to build, wire, test, and install the **AquaSentry Home** prototype.

AquaSentry Home is a dedicated surveillance node installed inside a domestic overhead water storage tank (e.g. Sintex 500L / 1000L). It monitors continuous water parameters (TDS, pH, Turbidity, Temperature), performs automated daily electrochemical Anodic Stripping Voltammetry (ASV) tests for trace dissolved Arsenic (As^{3+}), and runs the on-device **AquaNet-V1** neural network to provide real-time potability verdicts to your smartphone over Wi-Fi.

Workshop Tools & Consumables Required

Before starting, ensure you have the following basic tools:

Tool / Consumable	Recommended Specification	Purpose
Soldering Iron & Solder	40W–60W temperature-controlled, 60/40 leaded or lead-free solder	Making reliable electrical connections
Digital Multimeter (DMM)	Standard multimeter with DC voltage & continuity buzzer mode	Testing voltages and verifying zero short circuits
Wire Strippers & Cutters	20–30 AWG wire strippers, flush cutters	Trimming and stripping silicone hookup wire
Heat Shrink Tubing & Lighter / Heat Gun	1.5mm, 2.5mm, 4.0mm heat shrink sleeves	Insulating soldered splice joints against tank humidity
Screwdriver Set	Small Phillips (PH0, PH1) & 2.0mm / 2.5mm Allen keys	Securing M3 screws and PCB standoffs
Silicone Sealant / RTV	Neutral-cure clear silicone sealant (food-contact safe)	Hermetic waterproofing of cable gland entries
Breadboard / Perfboard	4cm x 6cm standard double-sided prototype perfboard	Soldering voltage divider resistors & op-amp circuit

Complete Bill of Materials (BOM) & Parts Checklist

Check off each item before beginning assembly:

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[ ] 1x ESP32 DevKit V1 (30-pin, Dual-Core 240MHz with Wi-Fi)
[ ] 1x ADS1115 16-Bit 4-Channel I2C ADC Module
[ ] 1x MCP6002 Rail-to-Rail Dual CMOS Op-Amp (DIP-8 or breakout module)
[ ] 1x Screen-Printed Gold 3-Electrode Strip (MAS-G1 or DropSens Gold WE, Ag/AgCl RE, Carbon CE)
[ ] 1x 12V DC Micro Peristaltic Dosing Pump (with silicone hose barbs)
[ ] 1x IRLZ44N Logic-Level N-Channel MOSFET (TO-220 package or breakout board)
[ ] 1x 1N4007 Diode (motor inductive flyback suppression)
[ ] 1x DS18B20 Waterproof Stainless Steel Temperature Probe (with 1-meter cable)
[ ] 1x Analog TDS / Electrical Conductivity Sensor Module (Gravity-style probe + board)
[ ] 1x Analog Optical Turbidity Sensor Module (IR sensor head + adapter board)
[ ] 1x Analog pH Sensor Kit (E-201-C glass composite probe + 4502C conditioning board)
[ ] 1x 12V 2A Regulated DC Power Adapter (wall plug with 5.5mm x 2.1mm DC barrel jack)
[ ] 1x LM2596 DC-DC Step-Down Buck Converter Module (with onboard adjustment trimmer)
[ ] 1x WS2812B NeoPixel RGB LED Ring (8 LEDs, 5V)
[ ] 1x Active 5V Piezo Buzzer
[ ] 2 Meters Food-Grade Silicone Tubing (2mm Inner Diameter / 4mm Outer Diameter)
[ ] Resistors: 1x 4.7kΩ (for DS18B20), 1x 10kΩ (MOSFET gate pull-down), 1x 10kΩ + 1x 20kΩ (Turbidity
[ ] 3D-Printed Parts: Enclosure Base & Lid, Rim Clamp Bracket, 15 mL Stripping Chamber (PETG)
[ ] Fasteners: 8x M3x10mm stainless screws, 4x M3 brass threaded heat-set inserts, 1x 70mm silicone
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Step 1: 3D Printing & Mechanical Preparation

AquaSentry Home lives in the high-humidity headspace of a domestic water tank. Material selection and print settings are critical:

1. **Filament Choice**:
2. **Print exclusively with PETG** (Food-contact safe, zero BPA, chemical-resistant to chlorine, heat resistant to 75°C).
3. *Do NOT use PLA* (PLA softens under rooftop solar heat at 55°C and degrades in high humidity).
4. **Slicer Settings**:
5. Layer Height: **0.20 mm**
6. Perimeters (Wall loops): **4 walls minimum** (essential for watertight layer bonding).
7. Infill: **40% Gyroid infill**.
8. Top & Bottom Layers: **5 solid layers**.
9. **Heat-Set Insert Installation**:
10. Using a soldering iron heated to 220°C, gently press the four **M3 brass threaded inserts** into the pre-modeled mounting posts of the main enclosure base until flush with the plastic rim.
11. **Fit the Silicone O-Ring**:
12. Stretch the 70mm silicone O-ring into the circumferential sealing groove on the upper rim of the main housing. Apply a very thin bead of silicone grease to preserve elasticity.

Step 2: Power Subsystem Calibration & Assembly

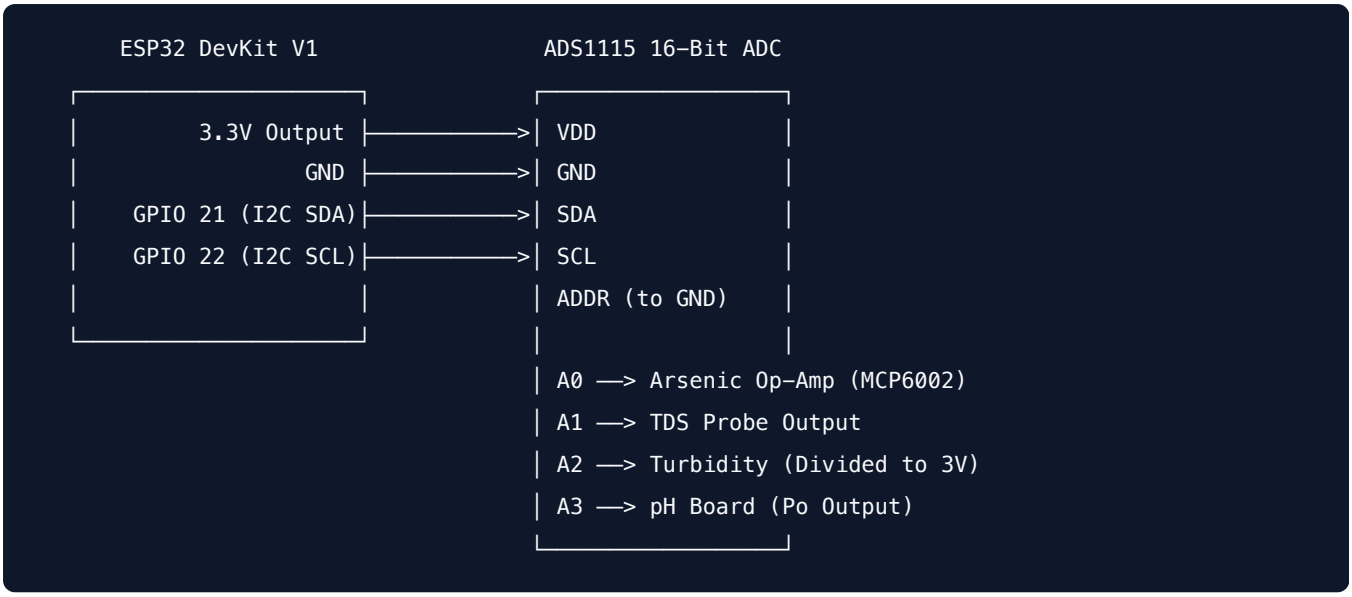
> **Caution:** > You must adjust the LM2596 step-down voltage with a multimeter **BEFORE** connecting the ESP32 or any sensor. Feeding raw 12V into the 5V pins will permanently destroy the electronics!

```
[12V 2A DC Jack] ──┬──────────────────────────────> [12V (+) Pump Power Rail]
                   │
                   └──> [LM2596 Buck Converter IN+]
                        Trimmer Pot tuned to 5.05V
                        [LM2596 Buck Converter OUT+] ──> [Clean 5V Rail for ESP32 & Sensors]
```

1. **Connect 12V Input**:
2. Solder the female DC barrel jack connector to the `IN+` and `IN−` terminals of the **LM2596 buck converter**.
3. Plug in the 12V 2A wall adapter.
4. **Voltage Calibration**:
5. Set your digital multimeter to **DC Volts (20V range)**.
6. Place the black probe on `OUT−` and the red probe on `OUT+` of the LM2596.
7. Using a small precision flathead screwdriver, slowly turn the brass screw on the blue potentiometer:
    - Counter-clockwise lowers the voltage.
    - Rotate until the multimeter reads exactly $5.05\ \mathrm{V}$ (this accounts for small voltage drops across wiring).
8. **Disconnect Power**:
9. Unplug the 12V adapter from the wall before proceeding.
10. **Create the Common Power Rails**:
11. Solder a 3-wire distribution bus on your mini perfboard:
    - **12V Rail (Yellow/Red wire)**: Feeds the peristaltic pump (+) terminal.
    - **5V Rail (Red wire)**: Feeds ESP32 `VIN`, 4502C pH board VCC, Turbidity board VCC, and NeoPixel VCC.
    - **Common Ground Rail (Black wire)**: Connect all subsystem grounds together (LM2596 `OUT−`, ESP32 `GND`, sensor grounds, and MOSFET source pin).

Step 3: Preparing the Analog Core & Precision ADC

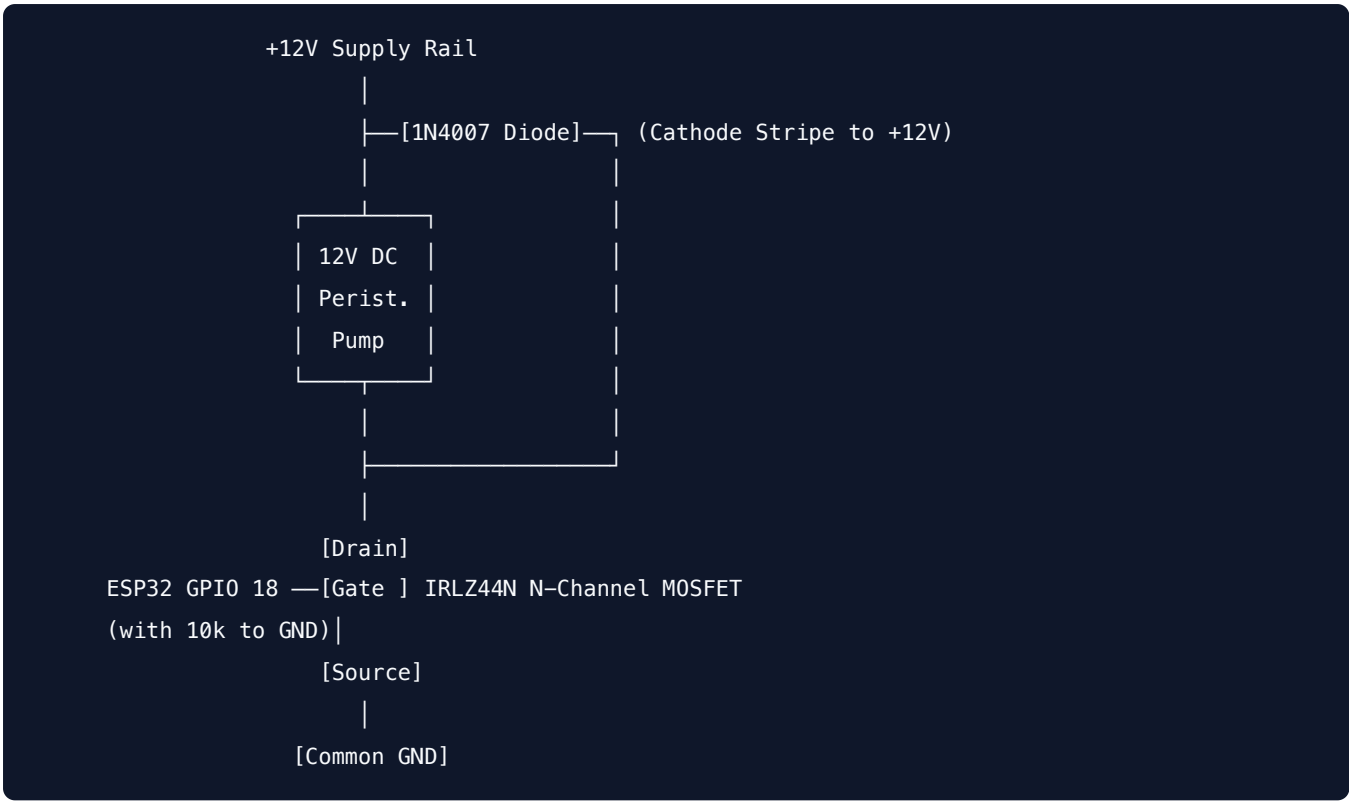
The ESP32's built-in ADC has non-linearity issues near 0V and 3.3V. The **ADS1115 module** provides precision 16-bit analog conversion across the entire range.



1. **Powering ADS1115**:
2. Solder a wire from the **ESP32's 3.3V pin** to the ADS1115 **VDD** pin.
3. *Why 3.3V?* This forces the ADS1115's I^2C communication lines to operate strictly at 3.3V, matching the ESP32's logic level with **zero logic level shifters needed**.
4. **Connect the I2C Bus**:
5. Solder ADS1115 **SDA** to ESP32 **GPIO 21**.
6. Solder ADS1115 **SCL** to ESP32 **GPIO 22**.
7. Solder ADS1115 **ADDR** pin to ADS1115 **GND** (this sets the default I^2C bus address to `0x48`).
8. **Assemble the MCP6002 Potentiostat Circuit**:
9. Mount the **MCP6002 dual rail-to-rail op-amp** on the mini perfboard.
10. Power MCP6002 V_{DD} (Pin 8) from ESP32 3.3V and V_{SS} (Pin 4) to Common GND.
11. Connect ESP32 **GPIO 25 (Internal DAC 1)** to the non-inverting input of Op-Amp 1 through a $10\text{k}\Omega/2.2\text{k}\Omega$ voltage divider. This allows the ESP32 to generate the Anodic Stripping Voltammetry potential ramp (-0.8V to $+0.4\text{V}$ relative to reference).
12. Connect the Working Electrode (WE) through the Transimpedance Amplifier (Op-Amp 2) to convert stripping nano-amp currents to micro-volts.
13. Connect the output of Op-Amp 2 to **ADS1115 Channel A0**.

Step 4: Arsenic Stripping Cell & Pump Fluidics

Leaving the gold electrode permanently submerged in domestic water causes biofouling within 48 hours. The peristaltic pump draws $15\ \text{mL}$ for testing once a day, then purges the chamber dry.



1. **Install Flyback Diode Across Pump Terminals**:
2. Solder a **1N4007 diode** directly across the pump's two DC terminals.
3. **Orientation**: The side of the diode with the silver stripe (cathode) **must connect to the positive (+12V) terminal**. This absorbs reverse inductive flyback spikes when the pump stops.
4. **Wire the IRLZ44N MOSFET Switch**:
5. Pin 1 (Gate): Solder a $10\ \mathrm{k}\Omega$ resistor between Gate and Common GND (pull-down). Solder Gate to ESP32 **GPIO 18**.
6. Pin 2 (Drain): Solder to the **negative ($-$) terminal** of the peristaltic pump.
7. Pin 3 (Source): Solder directly to the **Common GND** rail.
8. Connect the positive ($+$) terminal of the pump to the **+12V rail**.
9. **Plumbing the Silicone Tubing**:
10. Cut a 40 cm length of $2\ \mathrm{mm\ ID}/4\ \mathrm{mm\ OD}$ food-grade silicone tubing for the suction line.
11. Push one end onto the pump inlet barb. Push the other end through the bottom cable gland of the enclosure down into the water tank.
12. Connect a 10 cm piece of tubing from the pump outlet barb into the inlet port of the 3D-printed $15\ \mathrm{mL}$ stripping chamber.
13. Slide the **screen-printed gold electrode strip** into the chamber slot, ensuring the silicone gasket seals tightly against the electrode edges.
14. Connect a discharge tube from the chamber outlet port leading back to the tank drain.

Step 5: Auxiliary Water Quality Probes & Resistor Conditioning

Now wire the remaining 4 auxiliary sensors to provide complete geochemical context for AquaNet:

```
┌──────────────────────────────────────────────────────┐
| SENSOR WIRING & VOLTAGE CONDITIONING MATRIX           |
├──────────────────┬──────────────┬──────────┬─────────┤
| Sensor Probe     | Power (VCC)  | Ground   | Signal  |
├──────────────────┼──────────────┼──────────┼─────────┤
| DS18B20 Temp     | 3.3V (ESP32) | GND      | GPIO 4  |
| TDS Conductivity | 3.3V (ESP32) | GND      | ADS1115 A1
| Optical Turbidity| 5.0V (Buck)  | GND      | ADS1115 A2*
| pH Probe (4502C) | 5.0V (Buck)  | GND      | ADS1115 A3
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* Note: Turbidity signal passes through 10k/20k voltage divider!
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1. DS18B20 Temperature Probe

- Red Wire $\rightarrow$ ESP32 **3.3V**.
- Black Wire $\rightarrow$ Common **GND**.
- Yellow Wire (Data) $\rightarrow$ ESP32 **GPIO 4**.
- **Crucial Resistor**: Solder a $4.7\ \text{k}\Omega$ **resistor** between the Yellow Data wire and 3.3V (1-Wire pull-up).

2. Analog TDS Probe

- VCC $\rightarrow$ ESP32 **3.3V**.
- GND $\rightarrow$ Common **GND**.
- Analog Signal Pin $\rightarrow$ **ADS1115 Channel A1**.

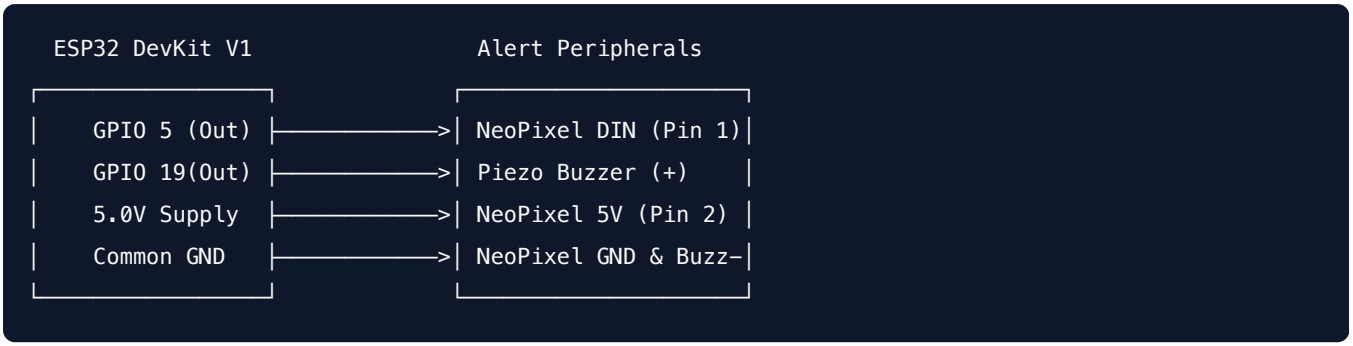
3. Optical Turbidity Sensor (Voltage Divider Conditioning)

- The optical turbidity board requires 5V VCC. In clear water, its output is $4.1\text{V} - 4.5\text{V}$, which would damage the 3.3V ADC.
- Solder the $10\ \text{k}\Omega$ **and** $20\ \text{k}\Omega$ **voltage divider** on your perfboard: `Turbidity Signal Pin ──>[ 10 kΩ ]─┬──> Connect to ADS1115 A2 (Max 3.0V) | [ 20 kΩ ] | GND`
- Connect VCC to the **5V rail**, GND to **Common GND**, and the center tap between the two resistors to **ADS1115 Channel A2**.

4. Analog pH Sensor Board (4502C)

- Connect VCC to the **5V rail** (pH glass probes require 5V reference amplifiers).
- Connect GND to **Common GND**.
- Connect the analog output pin (Po) directly to **ADS1115 Channel A3**.
- Connect the BNC connector of the glass electrode probe securely to the BNC jack on the board.

Step 6: Status Ring, Buzzer & Enclosure Assembly



1. **WS2812B NeoPixel 8-LED Ring**:
2. Mount the LED ring directly under the clear acrylic dome on the top lid.
3. Solder LED `+5V` to the **5V rail**.
4. Solder LED `GND` to **Common GND**.
5. Solder LED `DIN` to ESP32 **GPIO 5**.
6. **Active 5V Piezo Buzzer**:
7. Solder Buzzer $(+)$ pin to ESP32 **GPIO 19**.
8. Solder Buzzer $(-)$ pin to **Common GND**.
9. **Internal Wire Dress**:
10. Use small zip-ties to bundle the low-voltage sensor cables away from the 12V pump power wires to minimize electrical noise coupling into the ADC.
11. **Final Closure**:
12. Carefully position the electronics tray into the 3D-printed PETG base.
13. Place the clear top lid over the silicone O-ring gasket.
14. Tighten the four **M3x10mm stainless steel screws** in an alternating cross-pattern until the O-ring visibly compresses by approximately 30%.

Step 7: Benchtop Pre-Flight Check & Firmware Flashing

Before installing in the tank, perform this 5-minute electrical verification on your workbench:

1. The "Smoke-Test" (Continuity & Resistance Check)

- Set multimeter to **Continuity Buzzer mode**.
- Test between:
- 12V rail and GND: **Must NOT beep** (no short).
- 5V rail and GND: **Must NOT beep**.
- 3.3V rail and GND: **Must NOT beep**.
- If any rail beeps, find and fix the solder bridge before applying power!

2. Flashing the Firmware

- Connect the ESP32 to your computer using a Micro-USB or USB-C data cable.
- In the Arduino IDE or PlatformIO:
- Select Board: **ESP32 Dev Module** (CPU Frequency: 240MHz).
- Copy the verified ANSI C99 files: `aquanet_v1.h` and `aquanet_v1.c` into your project's `src/` directory.
- Enter your home 2.4 GHz Wi-Fi SSID and Password in `config.h`.
- Click **Upload**.
- Open the Serial Monitor at **115200 baud**. You should observe: `text [AquaSentry Home] Booting ESP32 Dual-Core Core... [AquaNet-V1] Initializing 23-node 112-synapse neural engine... OK. [ADS1115] Initializing 16-bit I2C ADC at 0x48... FOUND. [DS18B20] Dallas 1-Wire temperature sensor... FOUND (24.2 C). [Wi-Fi] Connected to Home Network. Local IP: 192.168.1.145 [mDNS] Web server responder active at http://aquasentry.local [NeoPixel] Self-test complete: Glowing Green.`

Step 8: Tank Installation & Daily Operation

1. **Mounting on the Water Tank**:
2. Open the overhead water tank inspection manhole (e.g. Sintex tank top lid).
3. Hook the 3D-printed PETG rim clamp over the tank lip and tighten the thumbscrew.
4. The main enclosure stays **outside/above the waterline** in the dome headspace.
5. **Submerging the Probe Harness**:
6. Lower the probe bracket (holding TDS, pH, Temp, and the silicone suction tube) into the water column until it rests **30–50 cm below the surface**.
7. *Ensure the suction tube tip is at least 15 cm above the tank bottom* to avoid pulling settled sediment into the sampling cell.
8. **Plugging In**:
9. Route the 12V DC power cable to the nearest electrical socket (typically the rooftop water pump starter switchboard or bathroom geyser socket).
10. **Accessing Your Dashboard**:
11. On your smartphone, tablet, or laptop connected to your home Wi-Fi, open your browser and navigate to: `text http://aquasentry.local`
12. *(Or type the ESP32's assigned IP address, e.g.* `http://192.168.1.145` *).*
13. You will see the real-time hydrogeochemical dashboard displaying:
    - **Arsenic**: Measured stripping concentration (mg/L).
    - **pH & TDS**: Acidity and mineral content.
    - **Turbidity**: Water clarity (alerts if tank cleaning is due).
    - **Neural Potability Status**:
    - 🟢 **Safe**: Arsenic $< 0.010\ \mathrm{mg/L}$, all parameters within WHO standards.
    - 🟡 **Caution**: Exceeds WHO guideline, within domestic limits.
    - 🔴 **Unsafe**: Exceeds BIS 10500 permissible limit ($> 0.050\ \mathrm{mg/L}$). Alarm sounds.

Troubleshooting Quick-Reference

Symptom	Probable Cause	Fix
ESP32 does not power on	LM2596 buck voltage not reaching VIN	Check DC adapter connection; verify LM2596 outputs 5.0V with multimeter.
Serial monitor reports `ADS1115 NOT FOUND`	Loose I2C wiring or incorrect VDD voltage	Verify SDA → GPIO 21, SCL → GPIO 22, and VDD is on 3.3V (not 5V).
Pump clicks but doesn't spin	Gate threshold not reached (IRF520 used instead of IRLZ44N)	Replace with a logic-level IRLZ44N MOSFET; verify 12V rail has power.
Temperature reads -127°C	1-Wire pull-up resistor missing or disconnected	Solder a $4.7\ \text{k}\Omega$ resistor between DS18B20 Data (Yellow) and 3.3V.
Arsenic stripping peak is noisy	Motor noise or ground loop coupling	Verify 1N4007 flyback diode is across pump; ensure common ground is solid.
Wi-Fi cannot connect	5 GHz network selected (ESP32 only supports 2.4 GHz)	Connect smartphone to router's 2.4 GHz band and enter credentials.

Congratulations!

Your **AquaSentry Home** autonomous surveillance node is now fully assembled, calibrated, and guarding your household drinking water with edge-grade neural precision.

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